Mingzhe Du

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Innovation 4.0, 3 Research Link, Singapore, 117602

Research Interests

Self-evolving Harness / Long-horizon Code Generation / Preference Alignment

Education

Nanyang Technological University — Singapore
PhD Candidate in Computer Science, College of Computing and Data Science — *Aug. 2022 – Present*
- Advisor: *Prof. Anh Tuan Luu*

The University of Melbourne — Melbourne, Australia
Master of Information Technology, Faculty of Engineering — *Oct. 2017 – Dec. 2019*
- Advisor: *Prof. Richard O. Sinnott*
- Specialization: Distributed Computing (Honorable Dissertation)

Central South University — Changsha, China
Bachelor of Information Security, School of Computer Science and Engineering — *Sep. 2013 – Jun. 2017*
- Excellent Scholarships *(GPA: 3.7/4.0)*

Professional Experience

TikTok, ***AI Innovation Center, Singapore*** — *May 2026 – Present*
- Research Scientist (Full Time / Soaring Star)
- Developing AI systems for large-scale code efficiency optimization, and bridging research and practice by deploying them to optimize production codebases.

National University of Singapore, ***Institute of Data Science*** — *Aug. 2021 – May 2026*
- Research Associate (Advisor: *Prof. See-Kiong Ng*)
- Affiliated with Cisco-NUS Accelerated Digital Economy Corporate Laboratory.

ByteDance, ***Tiktok Search, Beijing*** — *Dec. 2019 – Aug. 2021*
- Software Architect (Full Time)
- Architected a scalable, general-purpose search engine (*Search Eternal*) that now powers core search services for major ByteDance platforms, including *Douyin*, *TikTok* and *TouTiao*.

Academic Services

- **Conference Committee:** *NeurIPS (24,25,26)*, *ICLR (24,25,26,27)*, *ICML (25,26)*, *AISTATS (24,26,27)*, *AAAI (26,27)*, *CIKM (25,26)*, *AISI'26*, *ECAI'25*, *NAACL'24*, *EMNLP'24*, *ECAI'24*, *CLEF'23*.
- **Journal Reviewer:** *ACM Computing Survey*, *Frontiers in Artificial Intelligence*.
- **Volunteer:** *NeurIPS'25*, *ACL'25*, *WWW'24*, *EMNLP'24*, *AAAI'23*.
- **STEM Mentor:** *The University of Melbourne (2019-Present)*.

Selected Publications

[NeurIPS'25] Afterburner: Reinforcement Learning Facilitates Self-Improving Code Efficiency Optimization.
Mingzhe Du, *Luu Anh Tuan, Yue Liu, Yuhao Qing, Dong Huang, Xinyi He, Qian Liu, Zejun Ma, See-kiong Ng*

[ACL'25] CodeArena: A Collective Evaluation Platform for LLM Code Generation.
Mingzhe Du, *Luu Anh Tuan, Bin Ji, Xiaobao Wu, Dong Huang, Terry Yue Zhuo, Qian Liu, See-Kiong Ng*

[NeurIPS'24] Mercury: A Code Efficiency Benchmark for Code Large Language Models.
Mingzhe Du, *Luu Anh Tuan, Bin Ji, Qian Liu, See-Kiong Ng*

Publications

† *Corresponding Author,* ∗ *Equal Contribution.*

[**Preprint**] Mastermind: Strategy-Level Learning for Repository-Scale Vulnerability Reproduction.
Mingzhe Du, *Luu Anh Tuan, Dong Huang, Renyang Liu, Tianyi Wu, Zhijiang Guo, See-kiong Ng.*

[**Preprint**] PortBench: Measuring the Gap Between Correctness and Speed in CPU-to-GPU Code Generation.
Mingzhe Du, *Anh Tuan Luu, Dong Huang, See-Kiong Ng.*

[**Preprint**] Last Secure Code Benchmark: Is the Code You Vibe Secure?
Tianyi Wu, Chao Ge, Peng Chen, Fengji Zhang, Yongxing Wu, Yuchong Xie, Jian Li, Chaoyuan Peng, Kaikai Zhang, Dong Huang, ***Mingzhe Du***, *Yukai Huang, Jiaheng Zhang, Terry Yue Zhuo, See-Kiong Ng.*

[**Preprint**] Pareto-Gated Credit Assignment for Concise Unit-Test Generation with Reinforcement Learning.
Dong Huang, Mark Harman, Jie M. Zhang, Zhijiang Guo, ***Mingzhe Du***†, *See-Kiong Ng.*

[**Preprint**] MatReplace: A Reference-Free Benchmark for Material Replacement in Interior Scenes.
Mingzhe Du, *Thong Thanh Nguyen, Cong-Duy T Nguyen, See-Kiong Ng, Anh Tuan Luu.*

[**Preprint**] Measuring the Checker: Mutation Analysis for GPU-Kernel Benchmark Oracles.
Dong Huang, Tianyi Wu, Xuan Zhang, Renyang Liu, Luu Anh Tuan, See-Kiong Ng, ***Mingzhe Du***†.

[**Preprint**] Secure Code Generation via Online Reinforcement Learning with Vulnerability Reward Model.
Tianyi Wu, ***Mingzhe Du***, *Yue Liu, Chengran Yang, Terry Yue Zhuo, Jiaheng Zhang, See-Kiong Ng.*

[**NeurIPS'26**] Scaling Code LLM Training and Test-Time Inference via Execution-Free Reward Models.
Xiao Zhu, Xinyu Zhou, Boyu Zhu, Hanxu Hu, ***Mingzhe Du***, *Haotian Zhang, Huiming Wang, Zhijiang Guo.*

[**CIKM'26**] Paper Espresso: From Paper Overload to Research Insight. (Oral and Best Paper Nomination)
Mingzhe Du, *Luu Anh Tuan, Dong Huang, See-kiong Ng.*

[**ICML'26**] SWE-Perf: Can Language Models Optimize Code Performance on Real-World Repositories?
Xinyi He, Qian Liu, ***Mingzhe Du***, *Lin Yan, Zhijie Fan, Yiming Huang, Zejian Yuan, Zejun Ma.*

[**ICML'26 DL4C**] Nexus: Execution-Grounded Multi-Agent Test Oracle Synthesis
Dong Huang, ***Mingzhe Du***†, *Jie M. Zhang, Zheng Lin, Meng Luo, Qianru Zhang, See-kiong Ng.*

[**ACL'26**] Semantics-Aligned, Curriculum-Driven, and Reasoning-Enhanced Vulnerability Repair Framework (Oral and Best Paper Nomination)
Chengran Yang, Ting Zhang, Jinfeng Jiang, Xin Zhou, Haoye Tian, ***Mingzhe Du***, *Jieke Shi, Junkai Chen, Yikun Li, Eng Lieh Ouh, Lwin Khin Shar, David Lo.*

[**TOSEM'26**] Benchmarking LLMs for Unit Test Generation from Real-World Functions.
Dong Huang, Jie M. Zhang, Mark Harman, Qianru Zhang, ***Mingzhe Du***†, *See-Kiong Ng.*

[**EACL'26**] Pro-QuEST: Prompt-chaining Quiz Engine for testing Specialized Technical Product Knowledge
Sujatha Das Gollapalli, Mouad Hakam, ***Mingzhe Du***, *See-Kiong Ng, Mohammed Hamzeh.*

[**AMIYA'26**] Improving Arabic Dialectness in LLMs with Reinforcement Learning
Sujatha Das Gollapalli, Mouad Hakam, ***Mingzhe Du***, *See-Kiong Ng, Mohammed Hamzeh.*

[**ICLR'26**] Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts. (Oral)
Zhaomin Wu, ***Mingzhe Du***, *See-Kiong Ng, Bingsheng He.*

[**EMNLP'25**] On Assigning Product and Software Codes to Service Requests with Large Language Models.
Sujatha Das Gollapalli, Mouad Hakam, ***Mingzhe Du***, *See-Kiong Ng, Mohammed Hamzeh.*

[**NeurIPS'25**] EffiBench-X: A Multi-Language Benchmark for Measuring Efficiency of LLM-Generated Code.
Yuhao Qing∗*, Boyu Zhu*∗*,* ***Mingzhe Du***∗*, Zhijiang Guo, Terry Yue Zhuo, Qianru Zhang, Jie M. Zhang, Heming Cui, Siu-Ming Yiu, Dong Huang, See-Kiong Ng, Luu Anh Tuan.*

[**NeurIPS'25**] Guardreasoner-VL: Safeguarding VLMs via Reinforced Reasoning.
Yue Liu, Shengfang Zhai, ***Mingzhe Du***, *Yulin Chen, Tri Cao, Hongcheng Gao, Cheng Wang, Xinfeng Li, Kun Wang, Junfeng Fang, Jiaheng Zhang, Bryan Hooi.*

[**ICSE'25**] Measuring the Influence of Incorrect Code on Test Generation.
Dong Huang, Jie M. Zhang, Mark Harman, ***Mingzhe Du***†, *and Heming Cui.*

[ACL'25] AntiLeakBench: Preventing Data Contamination by Automatically Constructing Benchmarks with Updated Real-World Knowledge. (Oral and SAC Highlight Award)
*Xiaobao Wu, Liangming Pan, Yuxi Xie, Ruiwen Zhou, Shuai Zhao, Yubo Ma, **Mingzhe Du**, Rui Mao, Anh Tuan Luu, and William Yang Wang.*

[ICML'25] Current Model Licensing Practices are Dragging Us into a Quagmire of Legal Noncompliance. (Oral)
Moming Duan, **Mingzhe Du***, Rui Zhao, Mengying Wang, Yinghui Wu, Nigel Shadbolt, and Bingsheng He.*

[AAAI'25] Towards Verifiable Text Generation with Generative Agent. (Oral)
*Ji, Bin, Huijun Liu, **Mingzhe Du**, Shasha Li, Xiaodong Liu, Jun Ma, Jie Yu, and See-Kiong Ng.*

[SAC'25] Curriculum Demonstration Selection for In-Context Learning.
*Duc Anh Vu, Cong-Duy Nguyen, Xiaobao Wu, Nhat Hoang, **Mingzhe Du**, Thong Nguyen, and Anh Tuan Luu.*

[JMIR'25] Unraveling Online Mental Health Through the Lens of Early Maladaptive Schemas: AI-Enabled Content Analysis of Online Mental Health Communities.
*Beng Heng Ang, Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.*

[AAAI'24] From Static to Dynamic: Knowledge Metabolism for Large Language Models.
***Mingzhe Du**, Anh Tuan Luu, Bin Ji, and See-Kiong Ng.*

[AAAI'24] Chain-of-Thought Improves Text Generation with Citations in Large Language Models.
*Bin Ji, Huijun Liu, **Mingzhe Du**, and See-Kiong Ng.*

[ECAI'24] Counseling Responses for Mental Health Forum Questions with Maladaptive Schema Prediction.
*Sujatha Das Gollapalli, Beng Heng Ang, **Mingzhe Du**, and See-Kiong Ng.*

[TLDK'23] Constituency-Informed and Constituency-Constrained Extractive Question Answering with Heterogeneous Graph Transformer.
***Mingzhe Du**, Mouad Hakam, See-Kiong Ng, and Stéphane Bressan.*

[WWW'23] Identifying Checkworthy Cure Claims on Twitter.
*Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.*

[AAAI'23] Generating Reflective Questions for Engaging Gallery Visitors in ArtMuse.
*Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.*

[CLEF'22] NUS-IDS at CheckThat! 2022: Identifying Check-worthiness of Tweets using ChecktaT5.
***Mingzhe Du**, Sujatha Das Gollapalli, and See-Kiong Ng.*

Awards & Honors

- **NVIDIA Academic Grant**: *Predicting and Optimizing LLVM Compiler Pass Order in a Unified Model.*
- **Thinking Machines Lab Research Grant**: *Scaling Reinforcement Learning for Efficient Code Generation.*
- **NVIDIA Academic Grant**: *Scaling Laws of Reinforcement Learning for Code Efficiency Optimization.*
- **Staff Research Award** at *Institute of Data Science, National University of Singapore, 2025.*
- **Google Academic Research Award**: *SafeCodeX: Security-Aware Code Generation with LLMs, 2025.*
- **First Runner-Up Award** at the **Singapore Healthcare AI Datathon**: *Identifying ICU Patient with Central Lines at Risk of CLABSI, 2024.*
- **First Prize** at **Catalyst**: *A Path Recommendation System based on Data Aggregation and Analysis, 2019.*
- **First Prize** at **Codebrew**: *After Dark: Finding Safe and Fast Routes for Pedestrians at Night, 2019.*
- **Technology Pioneer Award** at the *Cybersecurity Talent Development and Entrepreneurship Forum, 2015.*
- **First Prize** at the *Mathematical Contest in Modeling*: *Study on the Wreckage Position of Malaysia Airlines Flight MH370, 2014.*

Miscellaneous

- **LinkedIn:** *https://www.linkedin.com/in/dumingzhe* **WeChat:** *spirit_song*
- **GitHub:** *https://www.github.com/elfsong* **Language:** *Mandarin (native) | English (professional)*

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